MCO-1D ORNITHOPTER

Bill of Materials — Revision 18

August 2026

✓ Prices are USD as of August 2026. Verify before ordering. All quantities assume one complete aircraft with one set of spares where noted. Software is open-source; no licence cost.

1. PROPULSION

Item	Qty	Source	Unit	Total	Notes
GoolRC 5010 360KV brushless motor	8	Amazon / GoolRC	~$18	~$144	1 per wing sector. Verify shaft size fits sector wheel.
ODrive S1 CAN controller	8	odriverobotics.com	~$79	~$632	1 per motor. node_id = wing_idx (0-7). CAN 1 Mbps.
2Ω 50W brake resistor	8	Amazon / Mouser	~$3	~$24	1 per S1 — CANNOT share. Pre-wired to S1 brake port.
30×2 mm CF sector wheel (custom)	8	Cut from CF sheet	—	—	100° arc, 7:1 effective gear ratio. Verify at TC-01 step C1.
Rope drive assembly (Dyneema 0.5mm)	8 sets	Online	~$5	~$40	Per-wing. Replace per RopeLifeTracker at 80% life.
4S LiPo 4000 mAh 35C	1	Hobbyking / Amazon	~$45	$45	Main power. Est. 420 g. 14.8 V nominal.
2S LiPo 1000 mAh	1	Hobbyking / Amazon	~$15	$15	Servo power — isolated from motor supply. Critical: F-013 gap.
60A ESC power distribution board	1	Amazon	~$12	$12	Feeds 8 × ODrive S1. 30A blade fuses per board.

2. SERVOS AND FEEDBACK

■ WARNING: All 16 positions (8 pitch + 8 sweep) use AGFRC IB53BHU. Order 18 units (16 installed + 2 spares). Programme ALL to 170° max travel via AGF-SPV3 before installation. Factory default is 180° which saturates MCP3208 ADC.

Item	Qty	Source	Unit	Total	Notes
AGFRC IB53BHU brushless servo	18	agf-rc.com or Amazon	$62.99	$1,134	36 kg·cm @ 7.4V, 444°/s, 69.5g, 180° travel, steel gears. 4th orange wire = analog feedback 300-3000 mV. 333 Hz PWM.
AGF-SPV3 servo programmer	1	agf-rc.com	~$25	$25	Required to programme max travel to 170°. Programme all 16 before installation. Also sets PWM rate to 333 Hz.
JST-SH 4-pin extension cable 30cm	16	Amazon	~$0.80	~$13	For 4th feedback wire routing to MCP3208 ADC boards.
Servo arm horn (M2 screw pattern)	24	Amazon	~$0.50	~$12	16 installed + 8 spare. Loctite 243 on all fasteners.

Torque margin — pitch	36 kg·cm vs 17.6 kg·cm required = 2.0× margin
Torque margin — sweep	36 kg·cm vs ~10 kg·cm required = 3.6× margin
Speed — pitch	444°/s vs 187°/s needed at 3 Hz = 2.4× margin
ADC range	300-3000 mV → 372-3723 counts (out of 4095) at 3.3V VREF
Max safe travel	170° (below ADC saturation at 180°)

3. FLIGHT COMPUTER AND SENSING

Item	Qty	Source	Unit	Total	Notes
Raspberry Pi 4B 4GB	1	Approved reseller	~$55	$55	Flight computer. Single point of failure — see F-007. Dual Pi needed for type certificate.
Pixhawk 4 Mini	1	Holybro	~$120	$120	MAVLink bridge, GPS, barometer, RC input. Pitot via MSP4525DO.
MS4525DO pitot sensor	1	Amazon / Holybro	~$25	$25	Forward of all wings, clean airflow. pitot_valid gates hover mode.
TF-Luna lidar rangefinder	1	Amazon	~$25	$25	0.1-8.0 m, I2C. Altitude hold, flare timing, airborne detection.
MPU-6050 IMU module	1	Amazon	~$5	$5	I2C addr 0x68. Attitude, rates, vibration RMS.
BMP280 barometer	1	Amazon	~$3	$3	I2C addr 0x76. Baro altitude fallback when lidar >8m.
DS18B20 temperature sensor	1	Amazon	~$2	$2	GPIO 4 (1-Wire). OAT for air density correction.
MCP3208 SPI ADC (8ch × 12-bit)	2	Digikey / Mouser	~$5	$10	#1 = pitch servo feedback (SPI1 CE1). #2 = sweep servo feedback (SPI1 CE0). VREF = 3.3V. Servo range: 372-3723 counts = 0-170°.
HX711 load cell amplifier	1	Amazon	~$3	$3	TC-01 static thrust measurement. GPIO 7 (CLK) + 26 (DT).
5 kg S-type load cell	1	Amazon	~$12	$12	Inline with tether at TC-01. Calibrate with 1 kg mass.
Waveshare 3.5" SPI touchscreen	1	Amazon / Waveshare	~$22	$22	Panel display. Optional — save 60g by running GS only on laptop.
2.4 GHz SBUS RC receiver	1	FrSky / Futaba compatible	~$30	$30	UART GPIO14/15. RC watchdog: 1 s loss → RTH.

4. CAN BUS AND INTERCONNECT

Item	Qty	Source	Unit	Total	Notes
MCP2515 CAN controller + TJA1050 transceiver	1	Amazon	~$5	$5	SPI0 CE0. CAN 1 Mbps. 120Ω termination at both ends of bus.
CAN bus cable 26 AWG twisted pair shielded	3m	Amazon	~$8	$8	Daisy-chain: Pi → S1#0 → S1#1 → ... → S1#7. Max 3m total.
120Ω CAN termination resistors	2	Mouser	~$0.10	$0.20	One at Pi end, one at S1#7. Required for signal integrity.
M2 nylon standoff set	1 pack	Amazon	~$8	$8	Pi and sensor board mounting. Non-conductive.
JST-XH 2-pin connectors assortment	1 pack	Amazon	~$8	$8	Motor power connections to ODrive.
22 AWG silicone wire red/black 10m	1 set	Amazon	~$12	$12	Power wiring. Silicone for flexibility.
XT30 connectors (M+F pairs)	6	Amazon	~$1	$6	LiPo → distribution board, board → ODrives.

5. AIRFRAME AND WING STRUCTURE

Item	Qty	Source	Unit	Total	Notes
10mm OD 0.5mm wall CF tube (wing rods)	2m	DragonPlate / Amazon	~$35	$35	Cut 8 rods: W1=772mm, W2=749mm, W3=735mm, W4=712mm port+stbd. Weigh each pair; note in build_measurements.py.
CF sheet 2mm (sector wheels, ribs)	200×300 mm	Amazon	~$15	$15	CNC or waterjet. Sector wheel: 100° arc, verified 7:1 ratio.
3K plain weave CF fabric (wing membrane)	0.5m²	Amazon	~$25	$25	S1223 profile. Turbulator tape at 33mm (upper) and 119mm (lower).
5-min epoxy	2 × 25ml	Hardware	~$8	$8	Wing layup and joints.
Turbulator tape 1mm wide	2m	Model aviation suppliers	~$5	$5	Upper: 33-58mm from LE. Lower: 119-132mm from LE.
Winglet sleeves (6-inch 4-slot PLA print)	8	3D print in-house	—	—	75-80° cant, 35° aft sweep. Verify D5 reach ≥85° stow.
Covert flap pivot pins M1.5	16	Amazon	~$0.20	$3	2 per wing: at 71mm and 111mm from LE. Spring-loaded return.
Aluminium fuselage tube 20mm OD	600mm	Metal supplier	~$10	$10	Central spine. CG: measure F1 from W1 pivot after assembly.
M3 titanium bolts + nyloc nuts assortment	1 pack	Amazon	~$12	$12	Wing pivot attachments. Nyloc only — no split washers on CF.
Foam vibration isolation pads (neoprene 3mm)	1 sheet	Amazon	~$8	$8	IMU and Pi mounting. Note: attenuates HF motor noise only; 1-3 Hz flapping will pass through — Kalman filter handles this.

6. COST SUMMARY

Section	Estimated USD
1. Propulsion (motors, ODrives, LiPos, ESCs)	~$912
2. Servos (18 × IB53BHU + programmer + accessories)	~$1,184
3. Flight computer, sensors, displays	~$280
4. CAN bus and interconnect	~$47
5. Airframe and wing structure	~$121
Contingency (10%)	~$254
TOTAL ESTIMATED BUILD COST	~$2,798

✓ Servos are the largest single line item at 42% of total. This reflects the decision to use instrumented brushless servos on all 16 positions rather than lighter analog alternatives. See servo selection rationale in Commissioning Manual Addendum A.

Budget driver	16 × AGFRC IB53BHU at $62.99 = $1,008 + programmer
Hardware gaps for type cert	F-007: 2nd Raspberry Pi ~$55. F-013: BEC/redundant servo supply ~$20.
Not included	Labour, workshop tools, test equipment, load cell fixture fabrication.

7. MASS BUDGET

Design MTOW: 3.448 kg. Estimated assembled: 5.095 kg. Aircraft hovers at 1.87 Hz with 2.58× T/W at estimated mass.

Component	Detail	Mass
Motors × 8	8 × ~85g	680g
ODrive S1 × 8	8 × ~85g	680g
IB53BHU servos × 16	16 × 69.5g	1,112g
Raspberry Pi 4B	1 × 46g	46g
Pixhawk 4 Mini	1 × 38g	38g
Main LiPo 4S 4000mAh	1 × ~420g	420g
Servo LiPo 2S 1000mAh	1 × ~75g	75g
Wing structure × 8	8 × ~40g est	320g
Fuselage / frame	est	400g
Electronics / wiring	est	200g
Sensors / display	est	80g
Fasteners / misc	est	44g
TOTAL ESTIMATED		4,095g

■ WARNING: Estimated total is 4.1 kg vs design MTOW 3.448 kg. Measure actual assembled mass (A1) before TC-01. Aircraft is confirmed flyable at 5.1 kg. Weight reduction: swap main LiPo to 2200 mAh (saves ~190g), remove GPS if not used (saves ~28g).

Design MTOW	3.448 kg (33.8 N)
Estimated actual	~4.1-5.1 kg (measure A1 in build_measurements.py)
Hover freq at 3.448 kg	1.54 Hz (estimated)
Hover freq at 5.095 kg	1.87 Hz (estimated) — well within 3.0 Hz cap
Thrust/weight at 3.0 Hz	2.32× (design) / 2.58× (estimated actual mass)